Degradation-Noise-Aware Deep Unfolding Transformer for Hyperspectral Image Denoising

Haijin Zeng, Kai Feng, Xudong Zhao, Jiezhang Cao, Shaoguang Huang, *Member, IEEE*, Hiep Luong, and Wilfried Philips

***Abstract*— Hyperspectral images (HSIs) play a pivotal role in fields, such as medical diagnosis and agriculture. However, it often contends with significant noise stemming from narrowband spectral filtering. Existing denoising techniques have their limitations: model-driven methods rely on manual priors and hyperparameters, while learning-based methods struggle to discern intrinsic noise patterns, as they require paired images with specific example noise for training, fail to capture critical noise distribution information, leading to unrobust denoising results. This work addresses the issue by presenting a degradation-noise-aware unfolding network (DNA-Net). Unlike training directly with the simulated noise, DNA-Net initially models general sparse and Gaussian noise through statistic distributions. It then explicitly represents image priors with a customized spectral transformer. The model is subsequently unfolded into an end-to-end (E2E) network, with hyperparameters adaptively estimated from noisy HSI and degradation models, effectively regulating each iteration. Furthermore, a novel U-shaped local–nonlocal–spectral transformer (U-LNSA) is introduced, simultaneously capturing spectral correlations, local features, and nonlocal dependencies. The integration of U-LNSA into DNA-Net establishes the first Transformer-based deep unfolding method for HSI denoising. Experimental results on synthetic and real noise validate DNA-Net's superior performance over state-of-the-art (SOTA) methods. Moreover, the DNA-Net, trained exclusively on mixed Gaussian noise and impulse noise, demonstrates the ability to generalize to unseen noise present in real images. Code and models will be released at: https://github.com/NavyZeng/DNA-Net.**

***Index Terms*— Noise model, remote sensing, spectral denoising, transformer, unfolding.**

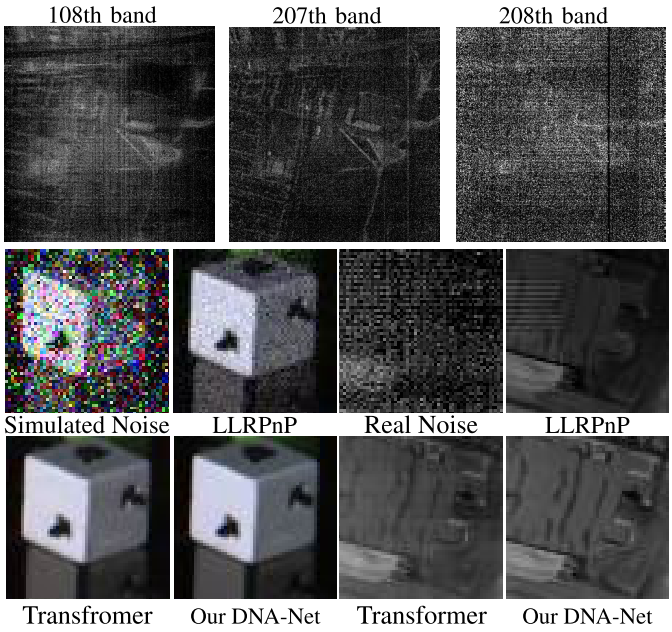


Fig. 1. We show the generalization problem of denoising networks. We train our DNA-Net, Transformer-based SERT [7] on Gaussian noise with $\sigma = 0.2$ and impulse noise with $p = 0.1$. When tested on the same noise, SERT shows outstanding performance. (Top) However, when applied to out-of-distribution noise, e.g., the mixture of various noises in real case. SERT suffers from a performance drop. DNA-Net maintains a reasonable denoising effect, despite also being trained on the Gaussian and impulse noise.

I. Introduction

HYPERSPECTRAL image (HSI) comprises numerous bands that cover a broad spectrum range. Due to its ability to incorporate wavelengths beyond the visible spectrum and distinguish subtle differences in various materials, HSI offers significant advantages for various applications, including remote sensing, face recognition, medical diagnosis, and classification [1], [2], [3], [4]. However, current hyperspectral imaging techniques always produce HSIs with substantial degradation, e.g., electronic noise and quantization noise [5], [6], [7], which emphasizes the importance of robust denoising algorithms to improve the image quality of HSI.

Research on HSI denoising can be categorized into four major directions: model-based models, plug-and-play (PnP) algorithms, end-to-end (E2E) deep neural networks, and deep unfolding methods. *Model-based methods* for HSI reconstruction rely on hand-crafted image priors such as sparsity-based total variation and spectral correlation-based low-rank representation and among others [8], [9]. These methods are theoretically grounded and interpretable. Recent advances have aimed to alleviate the need for manual parameter tuning, as demonstrated in works such as [10] and [11]. Despite these

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improvements, model-based approaches typically involve iterative optimization, which can increase the computation time required for convergence. PnP algorithms employ pretrained denoising network as PnP priors of traditional model-based methods [12], [13]. However, the performance of PnP methods is limited by the fixed nature of the pretrained networks, which cannot be retrained and the network is not customized. In addition, they also require parameters that need careful tuning. *E2E approaches* for HSI denoising usually use a convolutional neural network (CNN), to learn the mapping function from a noisy HSI to the clean HSI, which has the advantages of deep learning [7]. Nevertheless, they disregard the degradation mechanism and noise distributions of HSI by learning a brute-force mapping from the noisy HSIs to the underlying noise-free HSI. In addition, E2E methods lack theoretical foundations and interpretability.

Deep unfolding methods involve the utilization of a sequence of stages to map noisy HSI observation to clean one. Typically, each stage incorporates a fidelity phase, followed by a single-stage network that learns a denoiser prior [14], [15]. This presents a highly interpretable network architecture that explicitly characterizes both the image priors and the degradation model. These methods offer a hybrid approach that combines the strengths of both deep learning and model-based techniques, leading to promising denoising performance. However, current deep unfolding algorithms encounter two primary issues. First, the iterative learning process in these methods is highly dependent on the degradation pattern of the HSI yet current methods do not estimate the HSI degradation patterns and noise characteristics to adjust the denoising network in each iteration. Second, the existing deep unfolding methods predominantly employ CNNs, which exhibit limitations in capturing nonlocal self-similarity and long-range dependencies both of which are critical for HSI denoising.

Recently, the attention mechanism-based Transformer [16] model has emerged as a promising solution for overcoming the drawbacks of CNNs. The Transformer's strong capability to model the interactions of global contexts or nonlocal spatial regions has resulted in its wide application in image processing. For instance, the image classification [17], [18], object detection [19], [20], semantic segmentation [21], [22], human pose estimation [23], [24], and image restoration [25], [26]. However, using the Transformer in HSI denoising faces two primary issues. First, the computational complexity of the global Transformer [27] increases quadratically with spatial dimensions, rendering it unaffordable in some cases. Second, the receptive fields of local Transformer [17] are limited to position-specific windows, which hinders the matching of highly related content tokens during self-attention computation [28], and it also does not take into account the spectral correlation of HSI.

In this article, we propose a novel approach to address the problem of HSI denoising by designing a degradation-noise-aware unfolding network (DNA-Net). Unlike previous deep unfolding methods, DNA-Net accurately models general mixed noise using statistic distributions, to reduce uncertainty. Moreover, it adaptively estimates informative hyperparameters from the degraded noisy HSIs. These hyperparameters are then fed into each stage of DNA-Net to adaptively adjust the noise reduction.

In addition, a new U-shaped local–nonlocal–spectral transformer (U-LNSA) is proposed as the subdenoiser for each iteration. U-LNSA jointly extracts local contextual information, models nonlocal dependencies, and spectral correlation of HSI, while being computationally efficient. To achieve this, a spatial–spectral-split multihead self-attention (MSA) block is customized to form the basic unit of U-LNSA. Specifically, LNSA has three parallel attention branches, which calculate the self-attention within the local window, capture cross-window interactions by shuffling the tokens, and model the spectral information, respectively.

Finally, U-LNSA is integrated into DNA-Net to form an iterative architecture called a noise-aware unfolding denoising network. Experiments demonstrate that DNA-Net outperforms the latest state-of-the-art (SOTA) methods by over 1.2 dB. Our contributions are summarized as follows.

1) We propose a degradation-noise aware unfolding network for HSI denoising that accurately models the degradation pattern and noise distributions of HSI, to migrate the degradation uncertainty.
2) We propose a novel Transformer U-LNSA and plug it into DNA-Net to do HSI denoising. To the best of the authors' knowledge, DNA-Net is the first Transformer-based deep unfolding method for HSI denoising.
3) DNA-Net incorporates adaptive hyperparameter estimation, while achieving significant performance improvements over SOTA methods, particularly in scenarios involving heavy noise.

II. Related Works

A. Optimization and PnP-Based Methods

Optimization-based methods have been extensively employed for HSI denoising, with full-rank, low-rank matrix, and low-rank tensor approaches being the most commonly used techniques. These methods formulate the problem as an optimization objective that takes into account the unique properties of the spectrum and images. Various handcrafted regularization terms, such as total variation, wavelet constraint, and low-rank representation [29], [30], [31], [32], [33], [34], [35], [36], have been incorporated in the optimization framework to enhance the denoising performance.

These optimization-based methods are flexible enough to remove different types of noise [9], [37] and can be extended to tasks beyond denoising [38], [39], [40], [41]. However, the performance of these methods is limited by the degree of matching between the handcrafted regularization and the underlying properties of the HSI, especially for complex real scenes [9], [37]. To address this issue, PnP methods integrate optimization-based methods with a learning-based prior [13], [42], [43], [44], such as a PnP Gaussian denoiser, to handle complex noise types that cannot be easily modeled by handcrafted regularization [45], [46]. These methods can

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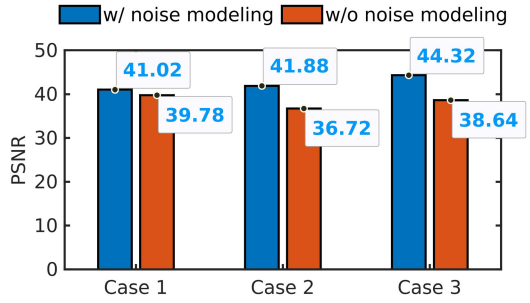


Fig. 2. Quantitative impact of noise modeling is examined on three distinct noise cases with mixed Gaussian noise and sparse noise: Case 1: $\mathcal{N}(0, 0.2)$, $p = 0.15$; Case 2: $\mathcal{N}(0, 0.2)$, $p = 0.05$; and Case 3: $\mathcal{N}(0, 0.2)$, where p is the percentage of the added sparse noise.

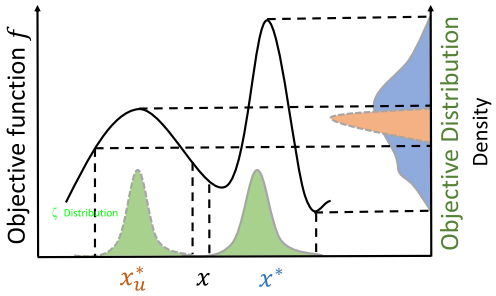


Fig. 3. Illustration of the effect on modeling noise uncertainty $\zeta \sim \mathcal{N}(0, \Sigma)$. For general linear model $f(x) = \Psi(x + \zeta)$, in denoising $\Psi = \mathbf{I}$, the input candidate x_u^* exhibits higher robustness than x^* as it leads a higher expectation (mean) and lower variance under uncertainty [67], [68], as shown in Fig. 2.

remove different types of noise and can also be extended to tasks beyond denoising, see [47], [48].

B. CNN and Unfolding Based Methods

Data-driven methods based on deep learning have also been employed for HSI denoising [14], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59]. In [60], a spatiospectral deep residual CNN was proposed that utilizes 3-D and 2-D convolutional filters to capture the dependencies of the images. The MemNet [61] network and one variation MemNetRes [62] (which is a combination of MemNet with the Hyres approach [59]), can provide competitive hyperspectral denoising results. Inspired by the success of the 2-D image denoising network DnCNN [49], Chang et al. [50] proposed HSI-DeNet, which learns multichannel 2-D filters to model spectral correlation. Yuan et al. [60] introduced a residual network structure with a sliding window strategy for remote sensed HSI. To further exploit spatial–spectral correlation, Dong et al. [51] designed a 3-D U-Net architecture. Despite successful applications of various methods to HSIs, most are limited to exploring interspectral correlations crucial for denoising. To tackle this limitation, the 3-D quasi-recurrent neural network (QRNN3D) [52] utilizes 3-D convolutional components and quasi-recurrent pooling to capture spatiospectral dependencies. In addition, Li et al. [7] propose a spectral enhanced rectangle transformer (SERT), which effectively leverages nonlocal spatial similarity and global spectral low-rank characteristic, being a new SOTA.

III. Proposed Method

A. Degradation-Noise Model of HSI

In the HSI imaging system, we denote the observed 3-D HSI and the latent noise-free HSI as tensor $\mathcal{Y} \in \mathbb{R}^{M\times N\times P}$ and $\mathcal{X} \in \mathbb{R}^{M\times N\times P}$, respectively, where P is the total number of spectral bands, and M and N denote the HSI's spatial height, width. Previous deep neural network-based denoising approaches mainly focus on learning an end-to-end deep model with paired training data, to reconstruct HSI $\hat{\mathcal{X}}$ by referencing the observed noisy image $\mathcal{Y}$. Mathematically, the end-to-end deep model is equivalent to solve the following problem:

$$\min_{\theta} \ \mathcal{T}(\hat{\mathcal{X}}, \mathcal{X}), \hat{\mathcal{X}} = \arg\min_{\mathcal{X}} \ F(\mathcal{X}, \mathcal{Y}) + \Phi_\theta(\mathcal{X}) \tag{1}$$

where $F(\mathcal{X}, \mathcal{Y})$ is image fidelity term and $\Phi(\mathcal{X})$ is the modeling of HSI prior. In this article, a deep neural network is designed to learn a data-adaptive function Φ, with θ being the network's trainable parameters. $\mathcal{T}(\cdot)$ denotes the loss function and θ is the learnable parameters of network. These methodologies, however, do not delineate noise characteristics to provide guidance during the learning phase. This deficiency arises from the fact that regression models, trained under supervision with a restricted set of noisy examples, do not factor in the inherent uncertainty associated with noise, e.g., the real noise in the top of Fig. 1 is hard to simulate. Consequently, they are limited in their ability to accurately identify and adapt to noise types that they have not been exposed to previously, as the comparison shown in the bottom of Fig. 1.

HSIs commonly suffer from various noise types such as Gaussian, sparse noise (e.g., impulse noise and stripes) and Poissonian noise [6], [9]. In modern hyperspectral sensors, increased spectral bands lead to a concern with heightened Poissonian noise due to reduced spectral bandwidth [63], [64], [65]. When photon counts exceed four, Poissonian noise can be approximated as nearly constant-variance additive Gaussian noise using variance-stabilizing transformations [66]. Hence, in this work, we assume the observation noise is additive Gaussian and sparse, and formulate the HSI degradation model with uncertainty accordingly

$$\mathcal{Y} = f(\mathcal{X}) = \Psi(\mathcal{X} + \zeta), \quad \zeta = \mathcal{N} + \mathcal{S} \tag{2}$$

where ζ denotes the noise degradation uncertainty, see Fig. 3 for an example. Symbol Ψ represents the imaging operator. Specifically, in the denoising context, it simplifies to $\Psi = \mathbf{I}$. Latent variables $\mathcal{N} \sim \mathcal{N}(0, \sigma_n^{-1})$ and $\mathcal{S} \sim \mathcal{L}(0, \lambda_s^{-1})$ represent the additive imaging Gaussian noise and the sparse noise on the observation, respectively. σ_n^{-1} and λ_s^{-1} are the hyperparameters with fixed value.

B. Unfolding Transformer and Noise Modeling

Based on noise uncertainty-guided degradation model (2), we then formulate a principled DNA-Net as depicted in

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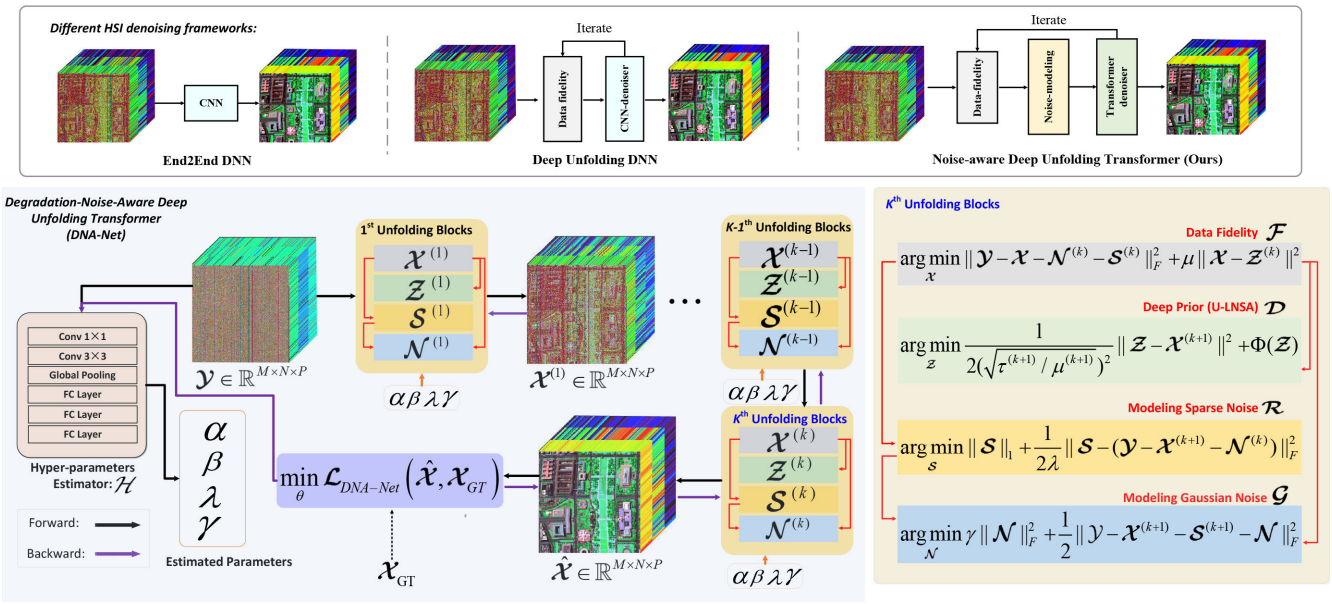


Fig. 4. Architecture of DNA-Net with K stages. The subproblems of $\mathcal{X}$, $\mathcal{S}$, and $\mathcal{N}$ have closed-form solutions, while the deep prior term $\mathcal{D}$ for $\mathcal{Z}$ is learned by the customized transformer U-LNSA. $\mathcal{H}$ estimates hyperparameters from noisy HSI $\mathcal{Y}$ and the degradation matrix $\mathbf{I}$. The learned parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, λ, and $\boldsymbol{\gamma}$ are fed into each stage of subsequent iterative learning ($\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are induced from $\boldsymbol{\mu}/\boldsymbol{\tau}$).

Fig. 4. DNA-Net starts by minimizing the following energy function:

$$\min_{\theta} \mathcal{T}(\hat{\mathcal{X}}(\mathcal{Y},\theta),\mathcal{X}), \quad \text{s.t., arg} \min_{\mathcal{X},\mathcal{N},\mathcal{S}} \frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{N}-\mathcal{S}\|_{\mathrm{F}}^2 + \tau\Phi(\mathcal{X}) + \lambda\|\mathcal{S}\|_1 + \gamma\|\mathcal{N}\|_{\mathrm{F}}^2 \quad (3)$$

where $\|\mathcal{Y}-\mathcal{X}-\mathcal{N}-\mathcal{S}\|^2$ is the data fidelity term, $\Phi(\mathcal{X})$ is the implicit image prior with a customized Transformer as shown in Section III-C, $\|\mathcal{S}\|_1$ $\|\mathcal{N}\|_{\mathrm{F}}^2$ are used to quantify general Sparse noise and Gaussian noise (especially for the case with heavy Gaussian noise) to reduce degradation uncertainty, and τ, λ, and γ are tradeoff hyperparameters.

1) Difference With SOTAs: It is noteworthy that our approach differs significantly from previous deep unfolding networks, such as the RGB image-based method proposed in [69], which primarily focuses on unfolding the maximum a posteriori (MAP) estimate using a data fidelity term and an implicit network regularizer. In contrast, our method targets HSI and addresses noise uncertainty by explicitly modeling the precise distributions of both general sparse noise and Gaussian noise, thereby achieving effective noise reduction in real-world scenarios. The efficacy of our noise modeling is demonstrated in Section IV-C and Fig. 2.

Furthermore, instead of employing a conventional CNN-based denoiser, we design a Transformer-based denoiser Φ specifically tailored for HSI. This transformer model adeptly captures both local and nonlocal spatial–spectral correlations inherent in HSIs. In addition, compared with traditional optimization approaches that incorporate noise modeling, our method uses trainable image prior modeling through neural networks, allowing it to capture complex image priors beyond the limited types, such as low-rank or sparse priors, used in conventional methods. In addition, the hyperparameters related to noise terms and image priors are adaptive and learnable, enabling our model to achieve an optimal balance between effective noise reduction and preservation of image details.

2) Alternating Optimization: By introducing an auxiliary variable $\mathcal{Z}$, and using half-quadratic splitting (HQS) algorithm for its simplicity and fast convergence, (3) is solved by minimizing the following iterative subproblems:

$$\begin{aligned}
\mathcal{X}^{(k+1)} &= \arg\min_{\mathcal{X}} \|\mathcal{Y}-\mathcal{X}-\mathcal{N}^{(k)}-\mathcal{S}^{(k)}\|_{\mathrm{F}}^2 + \mu\|\mathcal{X}-\mathcal{Z}^{(k)}\|^2 \\
\mathcal{Z}^{(k+1)} &= \arg\min_{\mathcal{Z}} \frac{\mu}{2}\|\mathcal{Z}-\mathcal{X}^{(k+1)}\|^2 + \tau\Phi(\mathcal{Z}) \\
\mathcal{S}^{(k+1)} &= \arg\min_{\mathcal{S}} \|\mathcal{S}\|_1 + \frac{1}{2\lambda}\left\|\mathcal{S}-\left(\mathcal{Y}-\mathcal{X}^{(k+1)}-\mathcal{N}^{(k)}\right)\right\|_{\mathrm{F}}^2 \\
\mathcal{N}^{(k+1)} &= \arg\min_{\mathcal{N}} \frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{N}-\mathcal{S}\|^2 + \gamma\|\mathcal{N}\|_{\mathrm{F}}^2 \qquad (4)
\end{aligned}$$

where $k = 0, 1, \ldots, K-1$ indexes the iteration.

1) *Update $\mathcal{X}$:* For the data fidelity term, i.e., $\mathcal{X}_{k+1}$ in (4), it is associated with a quadratic regularized least-squares problem, which has a closed-form solution

$$\mathcal{X}^{(k+1)} = \frac{\mathcal{Y}-\mathcal{N}^{(k)}-\mathcal{S}^{(k)}+\mu\mathcal{Z}^{(k)}}{1+\mu}. \quad (5)$$

2) *Update $\mathcal{Z}$:* By setting τ, μ as iteration-specific parameters and $\mathcal{Z}_{k+1}$ can be reformulated as follows:

$$\arg\min_{\mathcal{Z}} \frac{1}{2(\sqrt{\tau^{(k+1)}/\mu^{(k+1)}})^2}\|\mathcal{Z}-\mathcal{X}^{(k+1)}\|^2 + \Phi(\mathcal{Z}). \quad (6)$$

From the perspective of the Bayesian probability, (6) is equivalent to denoising image $\mathcal{X}^{(k+1)}$ with an Gaussian noise at level $(\tau^{(k+1)}/\mu^{(k+1)})^{1/2}$ [42]. Here, we customized a U-shaped transformer network, as shown in Fig. 5, to act as the denoiser $\mathcal{D}$, i.e.,

$$\mathcal{Z}^{(k+1)} = \mathcal{D}\left(\mathcal{X}^{(k+1)}, \sqrt{\tau^{(k+1)}/\mu^{(k+1)}}\right). \quad (7)$$

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3) *Update* $\mathcal{S}$: By using the soft-thresholding operator

$$R_\Delta(x) = \begin{cases} x - \Delta, & \text{if } x > \Delta \\ x + \Delta, & \text{if } x < \Delta \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $x \in \mathbb{R}$ and $\Delta > 0$, we can update $\mathcal{S}^{(k+1)}$ as follows:

$$\mathcal{S}^{(k+1)} = R_\lambda\left(\mathcal{Y} - \mathcal{X}^{(k+1)} - \mathcal{N}^{(k)}\right). \tag{9}$$

4) *Update* $\mathcal{N}$, it has a closed-form solution

$$\mathcal{N}^{(k+1)} = \frac{\mathcal{Y} - \mathcal{X}^{(k+1)} - \mathcal{S}^{(k+1)}}{1 + 2\gamma}. \tag{10}$$

3) Hyperparameters Estimation: The hyperparameters τ, λ, and γ in (7), (9), and (10) control noise reduction on the noisy input. However, their effectiveness depends on prior noise assumptions, which may not generalize well. To address this, we propose a data-driven approach using a hyperweight estimation network (Fig. 4). This captures the weighting principle objectively. In addition, μ in (4) is crucial for convergence. We automate its estimation based on iteration-specific HSI parameters, reducing manual tuning. Specifically, we denote the value of μ in the kth iteration as μ_k. Then, for estimating the parameters of (7), we set $(1/((\tau^{(k+1)}/\mu^{(k+1)})^{1/2})^2) = \mu^{(k+1)}/\tau^{(k+1)}$ as parameters to be estimated from HSI. Let $\alpha^{(k)} \overset{\text{def}}{=} \mu^{(k)}$, $\boldsymbol{\alpha} \overset{\text{def}}{=} [\alpha^{(1)}, \ldots, \alpha^{(k)}]$, $\beta^{(k)} \overset{\text{def}}{=} \mu^{(k)}/\tau^{(k)}$, and $\boldsymbol{\beta} \overset{\text{def}}{=} [\beta^{(1)}, \ldots, \beta^{(k)}]$, $\boldsymbol{\gamma} \overset{\text{def}}{=} [\gamma^{(1)}, \ldots, \gamma^{(k)}]$, $\lambda \overset{\text{def}}{=} [\lambda^{(1)}, \ldots, \lambda^{(k)}]$. Then, we can formulate our DNA-Net as an iterative scheme

$$\begin{aligned} (\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \lambda) &= \mathcal{H}(\mathcal{Y}, \mathbf{I}), \quad \mathcal{X}^{(k+1)} = \mathcal{F}\left(\mathcal{Y}, \mathcal{Z}^{(k)}, \alpha^{(k+1)}, \mathbf{I}\right) \\ \mathcal{Z}^{(k+1)} &= \mathcal{D}\left(\mathcal{X}^{(k+1)}, \beta^{(k+1)}\right) \\ \mathcal{S}^{(k+1)} &= \mathcal{R}\left(\mathcal{X}^{(k+1)}, \mathcal{N}^{(k)}, \lambda^{(k+1)}\right) \\ \mathcal{N}^{(k+1)} &= \mathcal{G}\left(\mathcal{X}^{(k+1)}, \mathcal{S}^{(k+1)}, \gamma^{(k+1)}\right) \end{aligned} \tag{11}$$

where $\mathcal{H}$ denotes the parameter estimator that takes the noisy observation $\mathcal{Y}$ and the identity degradation matrix $\mathbf{I}$ of the HSI system as inputs, $\mathcal{F}, \mathcal{R}$, and $\mathcal{G}$, equivalent to (5), (9), and (10), and $\mathcal{D}$ represents the denoiser in (7). The complete solving framework of DNA-Net is shown in Algorithm 1.

C. Customized HSI Transformer Denoiser

CNN-based denoisers face difficulties with long-range dependencies in HSI denoising. While Transformers can address this issue, their direct application to HSI denoising encounters challenges: limited receptive fields and high computational costs. HSI's high spectral resolution entails a strong spectral correlation, distinct from RGB. To address this, based on MSA [27], we propose a U-LNSA denoiser, leveraging local–nonlocal–spectral attention (LNSA) for (7).

1) Overview of Denoiser U-LNSA: The architecture of U-LNSA is depicted in Fig. 5(a) and is built in a U-Net style by utilizing the proposed LNSA block as its fundamental unit. Given an input $\mathcal{X}^{(k+1)} \in \mathbb{R}^{M\times N\times P}$, U-LNSA first applies a conv3 $\times$ 3 to map the concatenated reshaped $\mathcal{X}^{(k+1)}$ and stretched $\beta^{(k+1)}$ into feature $\mathcal{X}_0 \in \mathbb{R}^{M\times N\times C}$. Next, $\mathcal{X}_0$ is fed through the encoder, bottleneck, and decoder to be embedded into the deep feature $\mathcal{X}_d \in \mathbb{R}^{M\times N\times C}$. Each level of the encoder or decoder includes an LNSA and a resizing module. Fig. 5(b) shows that LNSA comprises two-layer normalization (LN) operations, an LNSA, and a feed-forward network (FFN) that is explained in Fig. 5(c). The downsampling and upsampling modules are implemented as strided conv4 $\times$ 4 and deconv2 $\times$ 2, respectively. Finally, a conv3 $\times$ 3 is applied to $\mathcal{X}_d$ to generate the residual image $\mathcal{T}^{(k+1)} \in \mathbb{R}^{M\times N\times P}$, and the output denoised image $\mathcal{Z}^{(k+1)}$ is obtained by adding reshaped $\mathcal{T}^{(k+1)}$ to $\mathcal{X}^{(k+1)}$: $\mathcal{Z}^{(k+1)} = \mathcal{T}^{(k+1)} + \mathcal{X}^{(k+1)}$.

2) Local–Nonlocal–Spectral Attention: As the basic unit of U-LNSA, the LNSA block is customized to model the local, and nonlocal dependencies and spectral correlation of HSI by using a memory-efficient group convolutional mechanism. As depicted in Figs. 5(b) and 6, LNSA first feed input into an LN block and a conv1 $\times$ 1 layer, and then passes it into the parallel local–nonlocal attention layer and the spectral attention block, respectively. Subsequently, the outputs of local attention and nonlocal attention are concatenated and fused by a conv1 $\times$ 1 layer, the fused outputs are concatenated with the output of spectral attention block and then fused by a conv1 $\times$ 1 layer. Finally, an LN and a feed-forward layer are employed to generate the output of LNSA.

3) Local–Nonlocal Attention Module: The input tokens of local–nonlocal-attention layer are denoted by $\mathcal{X}_{\text{in}} \in \mathbb{R}^{M\times N\times C}$. Subsequently, $\mathcal{X}_{\text{in}}$ is linearly projected into query $\mathcal{Q} \in \mathbb{R}^{M\times N\times C}$, key $\mathcal{K} \in \mathbb{R}^{M\times N\times C}$, and value $\mathcal{V} \in \mathbb{R}^{M\times N\times C}$ as follows:

$$\mathcal{Q} = \mathcal{X}_{\text{in}}\mathcal{W}^{\mathcal{Q}}, \quad \mathcal{K} = \mathcal{X}_{\text{in}}\mathcal{W}^{\mathcal{K}}, \quad \mathcal{V} = \mathcal{X}_{\text{in}}\mathcal{W}^{\mathcal{V}} \tag{12}$$

where $\mathcal{W}^{\mathcal{Q}}, \mathcal{W}^{\mathcal{K}}$, and $\mathcal{W}^{\mathcal{V}} \in \mathbb{R}^{C\times C}$ are the learnable parameters and biases are omitted for simplification. Then, the local contextual information and nonlocal dependencies are modeled by employing global MSA [27] and local window and shuffle-based MSA [17], [70]. The implementation of MSAs is parallelized to enhance efficiency. To this end, $\mathcal{Q}, \mathcal{K}$, and $\mathcal{V}$ are partitioned into two equal parts along the channel dimension

$$\mathcal{Q} = [\mathcal{Q}_1, \mathcal{Q}_2], \quad \mathcal{K} = [\mathcal{K}_1, \mathcal{K}_2], \quad \mathcal{V} = [\mathcal{V}_1, \mathcal{V}_2] \tag{13}$$

where $\mathcal{Q}_1, \mathcal{K}_1, \mathcal{V}_1 \in \mathbb{R}^{M\times N\times (C/2)}$ and $\mathcal{Q}_2, \mathcal{K}_2, \mathcal{V}_2 \in \mathbb{R}^{M\times N\times (C/2)}$.

Then, $\mathcal{Q}_1, \mathcal{K}_1$, and $\mathcal{V}_1$, and $\mathcal{Q}_2, \mathcal{K}_2$, and $\mathcal{V}_2$ are partitioned into nonoverlapping windows of size $p \times p$ and reshaped into $\mathbb{R}^{(MN/p^2)\times p^2\times (C/2)}$ first. However, in nonlocal branches, $\mathcal{Q}_2, \mathcal{K}_2$, and $\mathcal{V}_2$ are transposed from $\mathbb{R}^{(MN/p^2)\times p^2\times (C/2)}$ to $\mathbb{R}^{p^2\times (MN/p^2)\times (C/2)}$ to shuffle the positions of tokens and establish interwindow dependencies. Subsequently, $\mathcal{Q}_1, \mathcal{K}_1, \mathcal{V}_1$ and $\mathcal{Q}_2, \mathcal{K}_2, \mathcal{V}_2$ are split along the channelwise into h heads: $\mathcal{Q}_1 = [\mathcal{Q}_1^1, \ldots, \mathcal{Q}_1^h]$, $\mathcal{K}_1 = [\mathcal{K}_1^1, \ldots, \mathcal{K}_1^h]$, and $\mathcal{V}_1 = [\mathcal{V}_1^1, \ldots, \mathcal{V}_1^h]$. The dimension of each head is $d_h = (C/2h)$. $\mathcal{Q}_2, \mathcal{K}_2, \mathcal{V}_2$ are split into h heads: $\mathcal{Q}_2 = [\mathcal{Q}_2^1, \ldots, \mathcal{Q}_2^h]$, $\mathcal{K}_2 = [\mathcal{K}_2^1, \ldots, \mathcal{K}_2^h]$, and $\mathcal{V}_2 = [\mathcal{V}_2^1, \ldots, \mathcal{V}_2^h]$ The local and nonlocal self-attention $\mathcal{A}_1^i$ and $\mathcal{A}_2^i$ are calculated inside each head as follows:

$$\begin{aligned} \mathcal{A}_1^i &= \text{softmax}\left(\frac{\mathcal{Q}_1^i \, {\mathcal{K}_1^i}^{\mathsf{T}}}{\sqrt{d_h}} + \mathcal{P}_1^i\right)\mathcal{V}_1^i, \quad i = 1, \ldots, h \\ \mathcal{A}_2^i &= \text{softmax}\left(\frac{\mathcal{Q}_2^i \, {\mathcal{K}_2^i}^{\mathsf{T}}}{\sqrt{d_h}} + \mathcal{P}_2^i\right)\mathcal{V}_2^i, \quad i = 1, \ldots, h \end{aligned} \tag{14}$$

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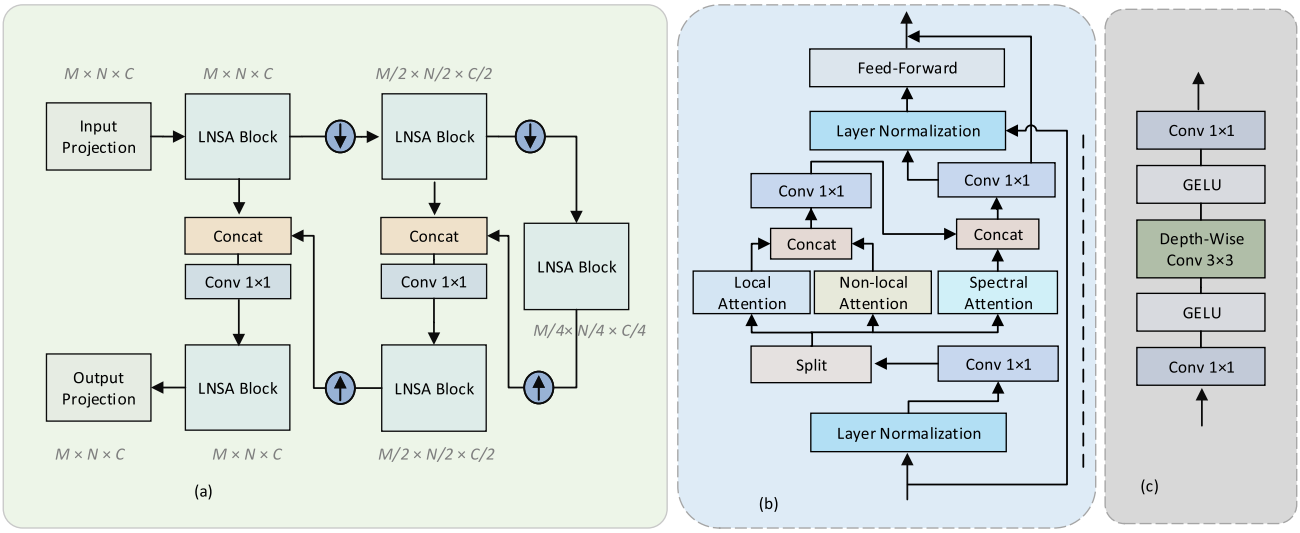


Fig. 5. Diagram of the proposed U-LNSA denoiser ($\mathcal{D}$) for DNA-Net, which is corresponding to subproblem (7), i.e., $\mathcal{D}$ in Fig. 4 and (11). (a) U-shaped U-LNSA, (b) LNSA block, and (c) FFN.

Algorithm 1 DNA-Net for HSI Denoising

Input: The observed noisy HSI tensor $\mathcal{Y} \in \mathbb{R}^{M\times N\times P}$, maximal iteration: K_{max}.

Output: The denoised tensor $\mathcal{X} \in \mathbb{R}^{M\times N\times P}$.

1: **Initialization:** The iteration $k = 0$, $\mathcal{X}^{(0)}$, $\mathcal{Z}^{(0)}$, $\mathcal{S}^{(0)}$, $\mathcal{N}^{(0)}$.
2: Estimate hyper-parameters $(\alpha, \beta, \lambda, \gamma)$ via $\mathcal{H}$;
3: **while** not converged and $k < K_{max}$ **do**
4: Update $\mathcal{X}^{t+1}$ via Eq. (6);
5: Update $\mathcal{Z}^{t+1}$ via Eq. (9);
6: Update $\mathcal{S}^{t+1}$ via Eq. (13);
7: Update $\mathcal{N}^{t+1}$ via Eq. (16);
8: Compute $\mathcal{T}(\cdot)$ (only for training)
9: **end while**

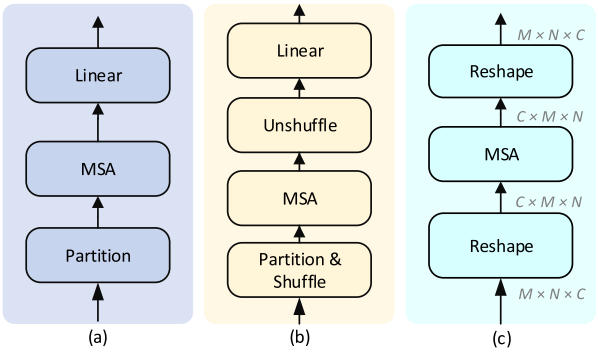


Fig. 6. Diagram of (a) local attention structure, (b) nonlocal attention block, and (c) spectral attention block, which is the key component of the LNSA block.

where $\mathcal{P}_1^i \in \mathbb{R}^{p^2\times p^2}$ and $\mathcal{P}_2^i \in \mathbb{R}^{(MN/p^2)\times(MN/p^2)}$ are the learnable parameters embedding the position information.

4) Spectral Prior Modeling Module: Meanwhile, a global spectral self-attention module (SM) is utilized in conjunction with the local–nonlocal branches to effectively capture the essential spectral correlations of HSI. Given an input $\mathcal{X}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$, by using SM block, the input is first transposed and reshaped into $\mathcal{X}^{\text{T}} \in \mathbb{R}^{C\times HW}$. Subsequently, X^{T} is linearly projected to $\mathcal{Q}^{\text{sm}} \in \mathbb{R}^{C\times HW}$, $\mathcal{K}^{\text{sm}} \in \mathbb{R}^{C\times HW}$, and $\mathcal{V}^{\text{sm}} \in \mathbb{R}^{C\times HW}$ as follows: $\mathcal{Q}^{\text{sm}} = \mathcal{W}_q^{\text{sm}}\mathcal{X}^{\text{T}}$, $\mathcal{K}^{\text{sm}} = \mathcal{W}_k^{\text{sm}}\mathcal{X}^{\text{T}}$, and $\mathcal{V}^{\text{sm}} = \mathcal{W}_v^{\text{sm}}\mathcal{X}^{\text{T}}$, where $\mathcal{W}_q, \mathcal{W}_k$, and $\mathcal{W}_v \in \mathbb{R}^{C\times C}$. Then, $\mathbf{Q}^{\text{sm}}, \mathbf{K}^{\text{sm}}$, and $\mathbf{V}^{\text{sm}}$ are split into N heads. The spectral attention of each head is computed as follows:

$$\mathcal{A}_j^{\text{sm}} = \mathcal{V}_j^{\text{sm}}\left(\text{Softmax}\left(\mathcal{K}_j^{smT}\mathcal{Q}_j^{\text{sm}}/\sqrt{d}\right)\right). \tag{15}$$

Subsequently, $\mathcal{A}_j^{\text{sm}}$ is concatenated in spectral dimension and projected to generate the output $\mathcal{X}_{\text{sm}}$.

Finally, $\mathcal{A}_2^i \in \mathbb{R}^{p^2\times(MN/p^2)\times d_h}$ is unshuffled by being transposed to shape $\mathbb{R}^{(MN/p^2)\times p^2\times d_h}$, and the final result of LNSA is produced by aggregating the outputs of local–nonlocal attention in (14) and spectral prior modeling block

$$\text{Conv } 1\times 1\left(\text{Concat}\left(\sum_{i=1}^{h}\mathcal{A}_1^i\mathcal{W}_1^i + \sum_{i=1}^{h}\mathcal{A}_2^i\mathcal{W}_2^i\right), \mathcal{X}_{\text{sm}}\right) \tag{16}$$

where $\mathcal{W}^i \in \mathbb{R}^{d_h\times C}$ is the learnable parameters.

IV. Experiments

A. Experimental Setup

We employ HSI consisting of 28 wavelengths ranging from 450 to 650 nm, obtained through spectral interpolation. The performance of the proposed method is evaluated through both simulation and real experiments. Simulation experiments are conducted using two datasets, namely CAVE [85] and KAIST [86]. The CAVE dataset comprised 32 HSIs with spatial dimensions of 512 × 512, while the KAIST dataset contained 30 HSIs with spatial dimensions of 2704 × 3376. The CAVE dataset is utilized as the training set, whereas ten scenes from the KAIST dataset are selected for testing purposes. For a fair comparison, all the tested models are trained from scratch with the same training set. To assess the general noise modeling capability, our model is exclusively trained using mixed Gaussian noise and impulse noise. The variations and percentages of these noises are randomly selected from the

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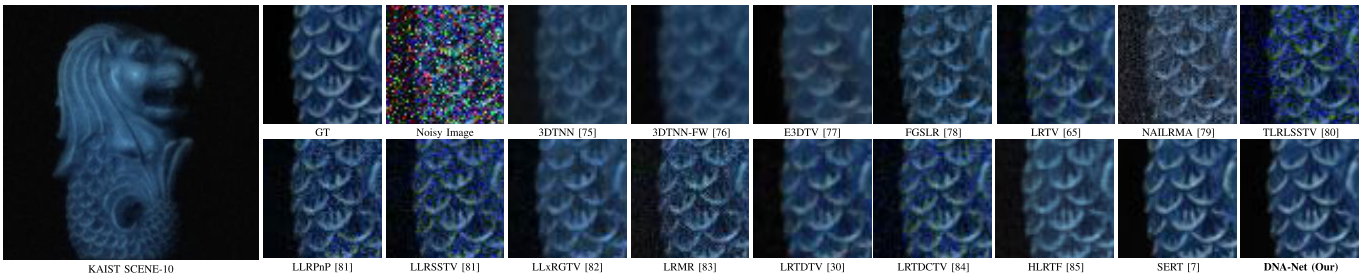


Fig. 7. Visual comparison of HSI denoising methods on KAIST dataset with simulated Gaussian and sparse noise.

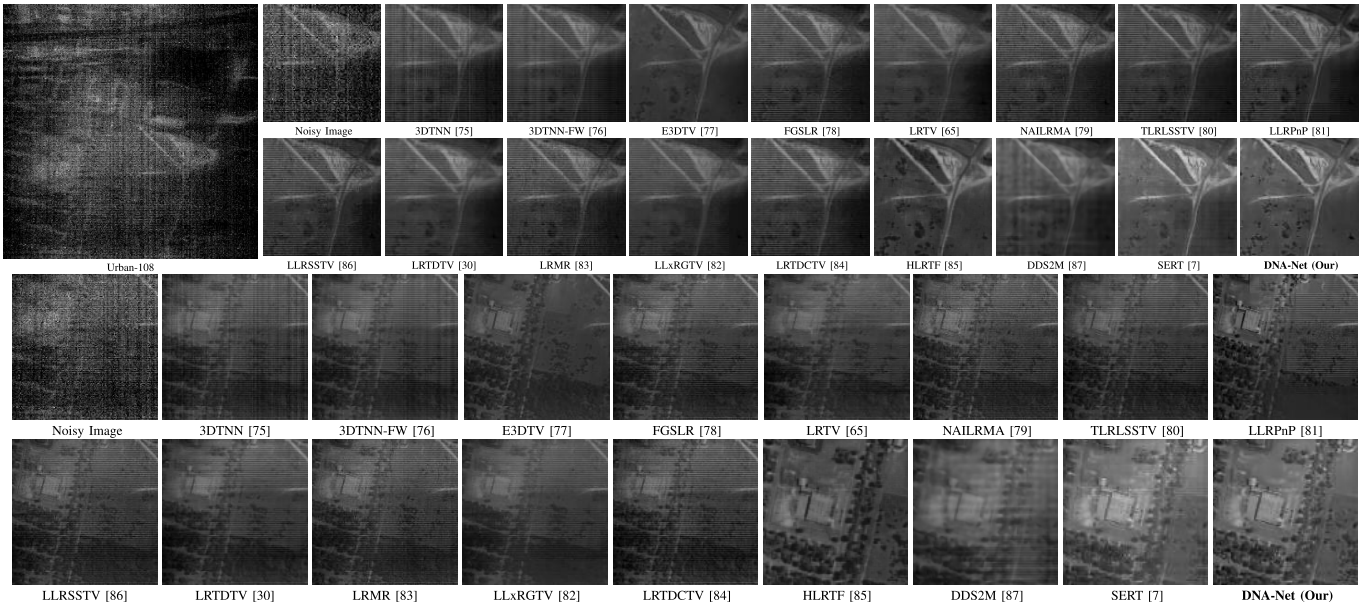


Fig. 8. Visual comparison of HSI denoising methods on urban affected by a range of noise sources including stripes, deadlines, atmospheric interference, water absorption, and other unidentified sources [9]. Here, DNA-Net is only trained with the Gaussian and impulse noise.

range of 0–0.2. Real experiments are conducted using Urban[1] which has real noise.

Pytorch is used to implement the proposed DNA-Net, which is trained with the Adam optimizer [87] and $\beta_1 = 0.9$ and $\beta_2 = 0.999$, for 110 epochs on a 2 × GTX 1080Ti 11 GB GPU using the cosine annealing scheme [88]. The initial learning rate is set to 4×10^{-4}. During both simulation and real experiments, training samples are randomly cropped from 3-D HSI cubes with 28 channels and spatial sizes of 256 × 256, utilizing a batch size of 2 with the basic channel $C = 28$ to store HSI information. The weights of Denoiser in different stages are unshared, and data augmentation techniques, such as random rotation and flipping, are used during training. In the training phase, the root mean square error (RMSE) is employed as the loss function $\mathcal{T}(\cdot)$ to quantify the discrepancy between the denoised HSIs and the ground truth. This choice is motivated by the focus of this work on noise and degradation modeling as well as network design.

B. Comparison With SOTA Methods

We compare the results of DNA-Net-5iter with five iterations, its light version DNA-Net-2iter with two iterations, and 18 SOTA methods including BM3D [45], LRTV [9]), NAILRMA [74], LRMR [78], NonLRMA [83], LRTDTV [30], LLRSSTV [81], TLR-LSSTV [75], LLxRGTV [77], 3DTNN [71], 3DTNN-FW [72], E3DTV [41], FGSLR [73], LRTDCTV [79], PnP-based LLRPnP [76], DDS2M [82] via diffusion, HLRTF [80] with tensor learning and transformer based SERT [7], and SSRT-UNet [84].

1) Simulation HSI Denoising:

Noise Cases (Training): The proposed models are initially trained using zero-mean Gaussian noise, with variance randomly selected from the range (0, 0.2), as well as salt-and-pepper noise, with the noise percentage also randomly chosen from the range (0, 0.2). The models are subsequently tested under the following four scenarios: *Case*-1: Gaussian noise $\mathcal{N}(0, 0.2)$, *Case*-2: Gaussian noise $\mathcal{N}(0, 0.2)$ + sparse (impulse) noise with $p = 0.05$, *Case*-3: Gaussian noise $\mathcal{N}(0, 0.2)$ + sparse noise with $p = 0.1$. *Case*-4: Gaussian noise $\mathcal{N}(0, 0.2)$ + sparse noise with $p = 0.15$.

Subsequently, to further evaluate the effect of stripe noise, in addition to Gaussian noise and salt-and-pepper noise, we trained the models with additional stripe noise. Specifically, 50% of the spectral bands were augmented with stripe noise, where the proportion of stripes relative to the image width

[1] http://www.tec.army.mil/hypercube/

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TABLE I

COMPARISONS UNDER NOISE (LEFT) *Case*-1 AND (RIGHT) *Case*-2. PSNR, SSIM, FSIM, ERGAS, AND RUNNING TIME ARE REPORTED

Method	Reference	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓	Time (s)
Noisy	None	16.094	0.110	0.426	807.880	none
LRTV	TGRS 2013	28.351	0.535	0.810	205.025	9.442
NAILRMA	TGRS 2017	23.215	0.407	0.822	356.335	8.632
LRMR	TGRS 2013	24.416	0.387	0.785	307.134	59.763
NonLRMA	TGRS 2017	27.897	0.725	0.883	204.716	16.043
LRTDTV	JSTAR 2017	30.147	0.618	0.861	159.869	43.858
LLRSSTV	JSTAR 2018	27.164	0.492	0.800	240.777	38.933
3DTNN-FW	TGRS 2019	27.754	0.538	0.850	209.240	22.227
TLR-LSSTV	TGRS 2021	27.981	0.539	0.821	218.898	100.600
LLRPnP	IA 2020	28.735	0.608	0.897	193.391	0.530
3DTNN	IS 2020	25.326	0.436	0.848	278.152	17.160
E3DTV	TIP 2020	28.996	0.854	0.878	189.818	9.288
LLxRGTV	SP 2021	28.915	0.572	0.881	184.659	39.750
FGSLR	TGRS 2021	30.333	0.668	0.901	157.146	192.628
FastHyMix	TNNLS 2021	**43.950**	0.754	0.964	51.982	0.318
LRTDCTV	JSTAR 2023	29.289	0.593	0.822	195.385	48.965
HLRTF	CVPR 2022	33.011	0.808	0.925	185.275	23
DDS2M	ICCV 2023	32.804	0.786	0.895	187.329	354
SERT	CVPR 2023	41.448	0.969	0.986	44.423	0.306
SSRT-Unet	TGRS 2024	41.515	0.968	0.983	45.648	0.322
DNA-Net-2iter	Ours	41.614	0.986	0.989	43.452	**0.082**
DNA-Net-5iter	Ours	42.332	**0.988**	**0.990**	**40.445**	0.205

Method	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓	Time (s)
Nosiy	13.515	0.056	0.292	982.660	none
LRTV	28.978	0.588	0.769	162.753	10.316
NAILRMA	21.765	0.359	0.710	384.629	9.010
LRMR	24.616	0.396	0.716	257.022	63.121
NonLRMA	20.626	0.429	0.774	447.370	17.475
LRTDTV	31.199	0.682	0.828	119.344	47.167
LLRSSTV	25.761	0.534	0.733	230.619	41.575
3DTNN-FW	29.123	0.810	0.895	151.480	25.897
TLRLSSTV	22.900	0.503	0.755	345.543	86.180
LLRPnP	27.206	0.634	0.807	189.142	0.583
3DTNN	26.138	0.703	0.898	225.699	22.766
E3DTV	32.634	0.907	0.942	168.679	11.667
LLxRGTV	30.657	0.732	0.886	128.963	46.797
FGSLR	31.617	0.727	0.860	115.760	285.600
FastHyMix	43.233	0.721	0.60	56.656	0.305
LRTDCTV	27.382	0.474	0.697	192.751	65.079
HLRTF	30.340	0.689	0.874	181.239	25
DDS2M	30.078	0.666	0.834	183.355	318
SERT	43.126	0.971	0.978	35.227	0.302
SSRT-Unet	43.224	0.976	0.982	36.332	0.261
DNA-Net-2iter	43.888	0.980	0.979	31.632	**0.091**
DNA-Net-5iter	**44.658**	**0.981**	**0.979**	**30.165**	0.214

TABLE II

COMPARISONS BETWEEN DNA-NET AND SOTA METHODS UNDER NOISE *Case*-3: GAUSSIAN NOISE $\mathcal{N}(0, 0.2)$ + SPARSE NOISE WITH $p = 0.1$

Method	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓	Time (s)
Noisy	11.830	0.045	0.331	1324.192	-
LRTV	28.872	0.617	0.806	225.460	9.302
NAILRMA	20.055	0.318	0.708	513.617	10.896
LRMR	24.204	0.400	0.745	317.549	60.537
NonLRMA	22.746	0.388	0.773	376.698	9.768
LRTDTV	30.629	0.695	0.848	154.044	43.426
LLRSSTV	26.010	0.510	0.754	288.215	40.266
SSTV	26.669	0.474	0.781	239.868	63.276
TLR-LSSTV	25.284	0.461	0.732	407.490	85.206
LLRPnP	27.275	0.643	0.832	243.813	0.748
LLxRGTV	29.296	0.705	0.890	176.948	45.746
3DTNN	26.739	0.671	0.863	237.061	20.755
3DTNN-FW	29.730	0.775	0.869	168.599	25.228
LRTDCTV	28.187	0.562	0.785	231.739	59.314
E3DTV	30.820	0.852	0.876	151.999	10.955
FGSLR	30.909	0.729	0.872	146.890	379.678
FastHyMix	39.941	0.750	0.954	74.075	0.308
SERT	42.350	0.969	0.984	45.665	0.293
DNA-Net-2iter (Ours)	43.072	0.981	0.988	42.095	**0.086**
DNA-Net-5iter (Ours)	**44.168**	**0.983**	**0.989**	**37.242**	0.212

TABLE III

COMPARISONS BETWEEN DNA-NET AND SOTA METHODS UNDER NOISE *Case*-4: GAUSSIAN NOISE $\mathcal{N}(0, 0.2)$ + SPARSE NOISE WITH $p = 0.15$

Method	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓	Time (s)
Noisy	10.688	0.034	0.247	1493.682	-
LRTV	27.957	0.535	0.729	217.996	9.498
NAILRMA	17.972	0.233	0.613	646.518	11.025
LRMR	22.666	0.355	0.669	372.022	60.654
NonLRMA	21.950	0.453	0.747	453.460	13.077
LRTDTV	30.086	0.637	0.788	161.100	47.716
LLRSSTV	23.405	0.546	0.700	361.783	40.168
SSTV	26.341	0.406	0.707	244.761	64.891
TLRLSSTV	26.008	0.585	0.763	286.956	87.575
LLRPnP	25.414	0.674	0.791	287.431	0.599
LLxRGTV	29.750	0.682	0.875	163.572	48.340
3DTNN	26.068	0.643	0.897	251.031	22.845
3DTNN-FW	28.953	0.759	0.882	179.162	28.159
LRTDCTV	25.592	0.398	0.644	279.517	67.074
E3DTV	30.343	0.879	0.915	189.866	11.735
FGSLR	29.914	0.632	0.804	160.725	491.061
FastHyMix	37.652	0.735	0.952	81.197	0.301
SERT	41.987	0.969	0.977	42.999	0.318
DNA-Net-2iter (Ours)	41.952	0.980	0.981	43.734	**0.086**
DNA-Net-5iter (Ours)	**43.780**	**0.983**	**0.984**	**35.252**	0.210

was randomly selected from the range (0.1, 0.2). For further details, please refer to our code. During testing, the models were evaluated on images containing stripe noise with the proportion of 0.15 relative to the image width (also 70% bands are added with stripe noise). FastHyMix [10] is added as comparison.

2) Evaluation Metrics: We employed the widely used quantitative picture quality indices (PQIs): PSNR, SSIM, FSIM [89], and ERGAS [90] metrics to evaluate the denoising performance quantitatively. Tables I–III depict the experiments on scenes 1–10 of the KAIST dataset, aims at removing mixed Gaussian and sparse noise. The proposed DNA-Net model is compared against SOTA denoising approaches. In noise *Case*-1, the DNA-Net-5iter model attains PSNR values of 42.33 and 44.66 dB, surpassing all other methods by an average of 1.2 dB. Moreover, with respect to SSIM, DNA-Net outperform the recent SOTA SERT [7] by a minimum of 0.1.

The visual analysis of the results is presented in Figs. 7 and 9–11, which demonstrate the efficacy of DNA-Net in reducing mixed noise while preserving the details. To evaluate if the enhanced performance is obtained at the expense of increased computational costs, we compare the PQIs results against computational cost, which are shown in Table I. The DNA-Net-2iter model delivers superior performance compared with most models, with the lowest computational cost, highlighting its efficiency and effectiveness.

3) Real HSI Denoising: The HYDICE Urban dataset has an original size of 207 × 207 × 210, with 189 bands remaining after excluding the water absorption band. Fig. 8 displays the grayscale images of the bands in the noisy HSI and the results after applying various restoration methods. Here, DNA-Net

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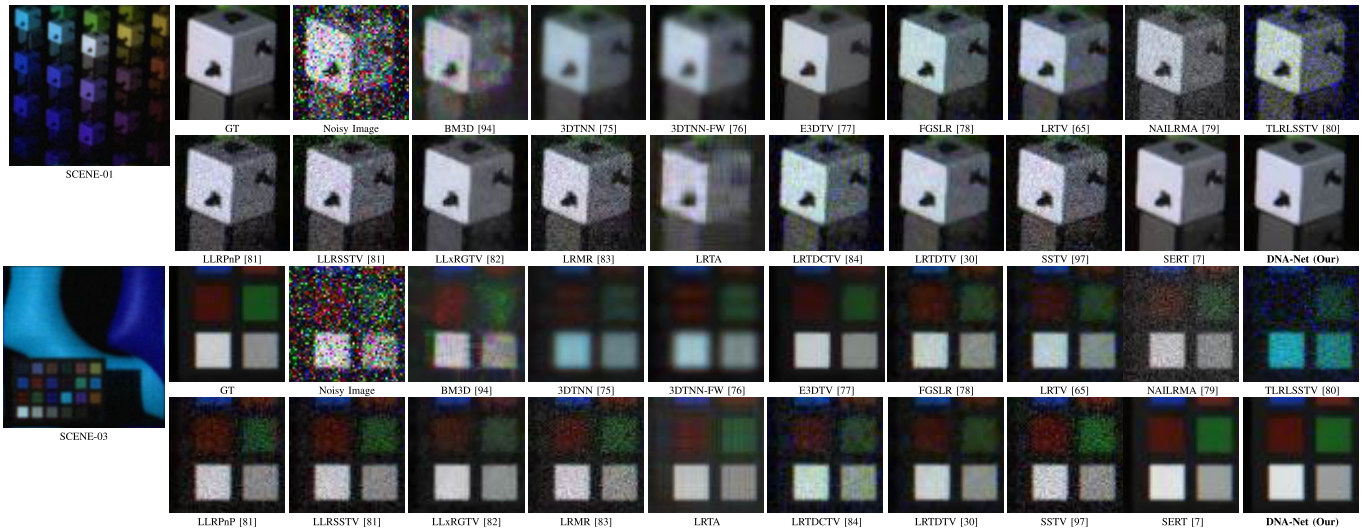


Fig. 9. Visual comparison of HSI denoising methods.

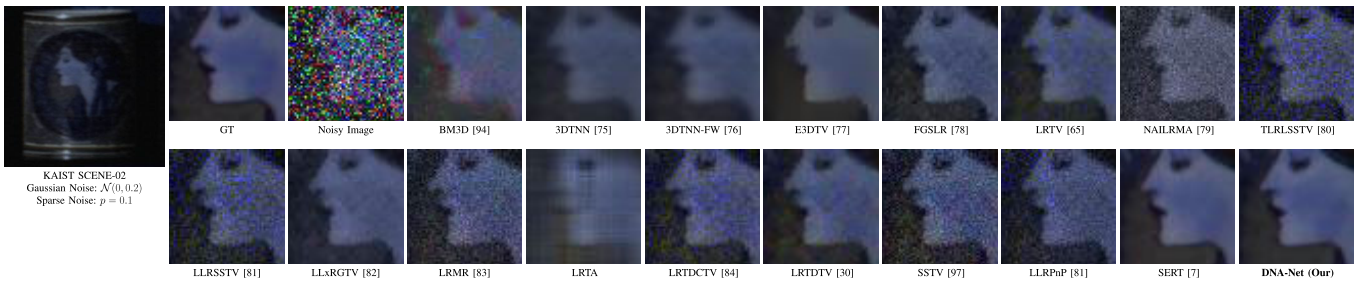


Fig. 10. Visual comparison of HSI denoising methods.

TABLE IV
PART OF THE MANUALLY TUNED HYPERPARAMETERS AND THE LEARNED HYPERPARAMETERS

Hyperparameters	α	β	λ
Manual-1	(2.6, 0.02, 0.002, 0.55, 23)	(0.004, 0.014, 1.5e-06, 2e-05)	(1e-07, 0.02, 1e-07, 0.02, 0.08)
Manual-2	(13, 0.1, 0.01, 2.75, 115)	(0.02, 0.07, 7.5e-06, 1e-04)	(5e-07, 0.1, 5e-07, 0.1, 0.4)
Manual-3	(39, 0.3, 0.03, 8.25, 345)	(0.06, 0.21, 2.2e-05, 3e-04)	(1.5e-06, 0.3, 1.5e-06, 0.3, 1.2)
Manual-4	(52, 0.4, 0.04, 11, 46)	(0.08, 0.28, 3e-06, 4e-05)	(2e-06, 0.4, 2e-06, 0.4, 1.6)
Learned	(26.40, 0.1734, 0.0243, 5.3798, 233.74)	(0.093, 0.1356, 0.0328, 1.68e-06, 2e-04)	(1e-06, 0.2179, 1e-06, 0.1925, 0.8125)

was trained exclusively on Gaussian and impulse noise, without exposure to stripe noise during training. Among the evaluated methods, DNA-Net demonstrated exceptional effectiveness in removing sparse noise, particularly stripe noise, and in preserving spatial details. Notably, DNA-Net achieved superior visual quality in the restored images, even when trained solely on the simulated Gaussian and impulse noise. This performance can be attributed to the proposed noise modeling term, which effectively separates unseen sparse noise.

C. Ablation Study

1) Hyperparameter Estimation and Initialization Discussion: In traditional iterative approaches involving noise modeling, manual tuning of noise term or image prior weights is a critical task. To eliminate the need for manual tuning and make our unrolling network independent of noise modeling hyperparameters, we introduce a subnetwork that learns these hyperparameters directly from the input noisy HSI and the degradation model.

To evaluate the efficacy of the hyperparameter learning approach, we conduct a comparative analysis between DNA-Net employing learned hyperparameters and several manually tuned hyperparameter groups. The actual number of manually tuned parameters and learned parameters is presented in Table IV. The manually tuned parameters encompass a range of different scales or are in close proximity to the learned parameters, indicating a search within the parameter space around the learned values using multiple scales.

The results showcased in Table VII highlight our findings, affirming that the automatic hyperparameter estimation approach not only alleviates the necessity for manual parameter tuning but also facilitates the acquisition of more optimal hyperparameters, leading to improved denoising performance. Furthermore, the results demonstrate that the unrolling network is also sensitive to hyperparameters, akin to traditional iterative approaches.

2) Effects of Attentions: Table V showcases outcomes from diverse component configurations. The initial row pertains to

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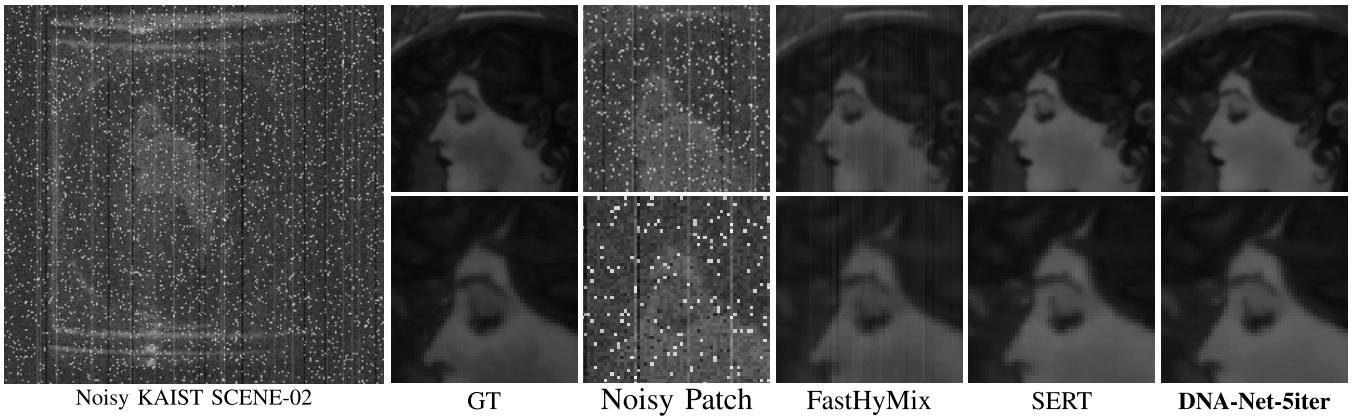


Fig. 11. Visual comparison of HSI denoising methods. Noise case: Gaussian noise: $\mathcal{N}(0, 0.2)$, sparse noise: $p = 0.1$, and stripe: 70% bands with the proportion of 0.15 relative to the image width.

TABLE V

ABLATION STUDY ON THE EFFECTS OF ATTENTIONS. PSNR, SSIM, FSIM, AND ERGAS ARE REPORTED

DNA-Net	Local Attention	Non-local Attention	Spectral Attention	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓
✓	✓			40.300	0.972	0.981	50.089
✓	✓		✓	41.247	0.978	0.986	45.116
✓	✓	✓		40.563	0.973	0.985	49.236
✓		✓	✓	40.721	0.976	0.985	47.313
✓	✓	✓	✓	**41.489**	**0.986**	**0.988**	**44.106**

TABLE VI

ABLATION STUDY ON NOISE MODELING: REMOVING THE NOISE MODELING TERM AND RETRAINING THE MODEL

Input noise	Noise modelling	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓
$p = 0.15$	W/	**41.023**	**0.984**	**0.987**	**47.021**
$\mathcal{N}(0, 0.2)$	W/O	36.721	0.954	0.973	77.601
$\mathcal{N}(0, 0.2)$	W/	**44.316**	**0.987**	**0.990**	**40.572**
	W/O	39.783	0.979	0.987	55.480
$p = 0.05$	W/	**41.881**	**0.987**	**0.989**	**42.636**
$\mathcal{N}(0, 0.2)$	W/O	38.642	0.975	0.983	63.159

TABLE VII

ABLATION STUDY ON THE HYPERPARAMETER ESTIMATION

Group $(\alpha, \beta, \lambda, \gamma)$	PSNR ↑	SSIM ↑	FSIM ↑	ERGAS ↓
Manual-1	23.69	0.549	0.927	261.6
Manual-2	33.71	0.854	0.983	93.82
Manual-3	39.67	0.942	0.985	50.90
Manual-4	31.54	0.809	0.954	133.6
Learned	**44.65**	**0.981**	**0.979**	**30.17**

DNA-Net employing exclusively local attention within the spatial domain. Rows 2–5 encompass DNA-Net variations integrating spectral attention, nonlocal attention, and the fusion of local, spectral with nonlocal attention. The findings underscore that the inclusion of spectral and nonlocal attention modules, enhancing the capture of spatial–spectral details, yields substantial enhancements in denoising performance with respect to PSNR and SSIM.

3) Noise Modeling: We then conduct the ablation study on noise modeling, which enhances the understanding of the proposed approach. Table VI presents the results, illustrating the impact of noise modeling on reducing both the Gaussian noise and Sparse noise. Interestingly, the analysis reveals that the improvement in reducing *sparse* noise is more pronounced compared to *Gaussian* noise. This observation can be attributed to several factors. First, the data fidelity term with l_2 norm exhibits a higher capability to mitigate Gaussian noise. In addition, it is possible that the network without noise modeling can also reduce the Gaussian noise to a certain extent due to its inherent characteristics.

4) Initialization: Since the input is the noisy HSI captured by cameras, and the task is to reduce noise, no initialization is needed for the input or output. The learnable hyperparameters are initialized with random samples from a Gaussian distribution. Other hyperparameters, such as batch size and learning rate, are manually fine-tuned during training to achieve optimal PSNR given the available GPU resources.

V. CONCLUSION

This article addresses three issues that arise in previous model-based or learning-based HSI denoising methods. These issues include a lack of informative parameter estimation for iterative learning, the inability to model noise distributions, and limitations in capturing long-range dependencies due to their primary reliance on CNN-based approaches. To overcome these challenges, we propose a new MAP-based unfolding framework called DNA-Net. The DNA-Net framework is capable of effectively modeling both sparse and Gaussian noise distributions and accurately estimating parameters from a noisy HSI and degradation matrix. Furthermore, we introduce a new Transformer block U-LNSA and integrate it into the DNA-Net framework. Experimental results indicate that our DNA-Net outperforms SOTA while showing robust performance on real noise.

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